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The power density indicator calculation (D_P) for the scheme for PR3 operation is—

$$D_P = \frac{P}{\sum_{i=1}^{n}\left(\bar{E}_i \cdot A_i\right)} \qquad \text{F4}$$

where

D_p = power density indicator, in watts per lux per square metre ($W.lx^{-1}.m^{-2}$)

P = system power = 470 W

$\bar{E}_i$ = calculated maintained average horizontal illuminance is 1.79 lx (say) (for subcategory PR3)

A_i = size of the sub-area (1000 × 18) = 18 000 m^2

n = number of sub-areas to be lit = 1

Therefore the PR3 state power density indicator = 470/(1.79 × 18 000) = **0.01458 $W.lx^{-1}.m^{-2}$**.

The power density indicator calculation (D_P) for the scheme for PR5 operation is—

For PR5 state (required level is 0.85 lx) the dim level is approximately half compared with PR3 state (required level is 1.75 lx), thus for LED luminaires the luminaire power will also reduce by a factor of 0.5, to 10 W per unit.

The system power (P) calculation for the above example is—

$$P = \sum_{k=1}^{n_{lp}} P_k + P_{ad} \qquad \text{F5}$$

where

P = total system power of the lighting scheme (W)

P_k = operational power of light point = (10 + 2) = 12 W

P_{ad} = total operational power of system devices = 30 W

Therefore the system power $(P) = (20 \times 12) + (30) = 270\,W$.

The power density indicator calculation (D_P) for the scheme for PR5 operation is—

$$D_P = \frac{P}{\sum_{i=1}^{n}\left(\bar{E}_i \cdot A_i\right)} \qquad \text{F6}$$

where

D_P = power density indicator, in watts per lux per square metre ($W.lx^{-1}.m^{-2}$)

P = system power = **270 W**

$\bar{E}_i$ = maintained average horizontal illuminance is 0.89 lx (say) (for subcategory PR5)

A_i = size of the sub-area (1000 × 18) = 18 000 m^2

n = number of sub-areas to be lit = 1

Therefore the PR5 state power density indicator = 270/(0.89 × 18 000) = **0.01685 $W.lx^{-1}.m^{-2}$**.

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